



Operations Research

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Solution to the Training Exercises, Sheet 11

Training Exercises

Exercise T11.1. *Convex Functions - Examples*

Problem

Check whether the following functions are convex.

- a) $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^4 - x$
- b) $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = |x| + \cos(x)$
- c) $\max\{f(x), g(x)\}$ for $f, g : \mathbb{R}^n \rightarrow \mathbb{R}$ convex
- d) $f : \mathbb{R}^n \rightarrow \mathbb{R},$

$$f(x) = \begin{cases} \|x\| & \text{if } \|x\| \leq 1 \\ \|x\|^2 & \text{else} \end{cases}$$

Solution

- a) f is convex, since $f''(x) = 12x^2 \geq 0$ for all $x \in \mathbb{R}$. Therefore, f is also quasi-convex.
- b) f is not convex: Set $\bar{x} = \pi$ and $\bar{y} = 3\pi$. It holds that

$$f\left(\frac{1}{2}\bar{x} + \frac{1}{2}\bar{y}\right) = f(2\pi) = 2\pi + 1 > 2\pi - 1 = \frac{1}{2}f(\bar{x}) + \frac{1}{2}f(\bar{y}).$$

- c) For f, g convex, we have

$$\begin{aligned} \max\{f(\lambda x + (1-\lambda)y), g(\lambda x + (1-\lambda)y)\} &\leq \\ \max\{\lambda f(x) + (1-\lambda)f(y), \lambda g(x) + (1-\lambda)g(y)\} &\leq \\ \lambda \max\{f(x), g(x)\} + (1-\lambda) \max\{f(y), g(y)\} \end{aligned}$$

Hence, $x \mapsto \max\{f(x), g(x)\}$ is convex.

- d) f can also be written as $f(x) = \max\{\|x\|, \|x\|^2\}$. $\|x\|$ is convex, which follows from the triangle inequality:

$$\|\lambda x + (1-\lambda)y\| \leq \lambda\|x\| + (1-\lambda)\|y\|$$

$\|x\|^2$ is also convex. This can be seen, for example, from the fact that the Hessian matrix is $2I$, where I denotes the identity matrix. Obviously, all eigenvalues are positive, from which convexity follows according to the lecture. Therefore, from part c, it follows that f is convex.

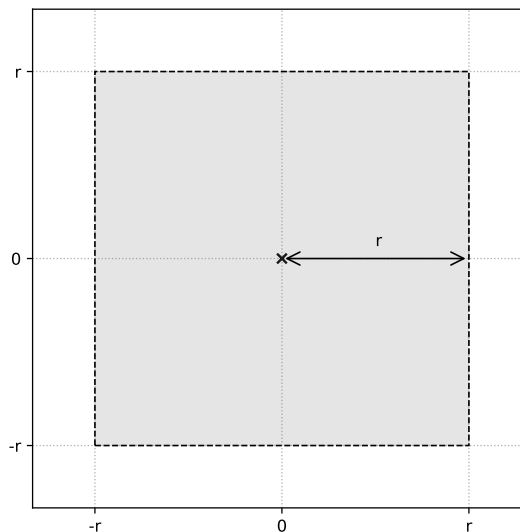
Exercise T11.2. Convex functions

Problem

Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$,

$$f(x, y) = \exp(2x) + xy + y^2.$$

For $r > 0$, denote with $Q_r = \{(x, y) \in \mathbb{R}^2 : \max(|x|, |y|) < r\}$ the square around $(0, 0)$ with side length $2r$, (see Figure below).



Give an $r > 0$ such that $f : Q_r \rightarrow \mathbb{R}$ is convex. Justify your answer.

Solution

The Hessian matrix is given by

$$\begin{pmatrix} 4\exp(2x) & 1 \\ 1 & 2 \end{pmatrix},$$

with leading 1×1 principal minor $4\exp(2x)$ and 2×2 principal minor $8\exp(2x) - 1$. The matrix is positive definite exactly when $8\exp(2x) - 1 > 0 \Leftrightarrow x > \ln(1/8)/2$. We choose $r = |\ln(1/8)/2|$. Then, since $\ln(1/8) < 0$, it holds that the Hessian matrix is positive definite for all $(x, y) \in Q_r$ (and hence, in particular, positive semidefinite). Therefore f is convex on Q_r . Of course, this is also true for $Q_{r'}$ with $0 < r' < r$.

We additionally show that r is maximal. So let $r' > r$. Since $\max(|\ln(1/8)/2 - (r' - r)/2|, |0|) = -\ln(1/8)/2 + (r' - r)/2 = r + (r' - r)/2 < r'$, $(x, y) = (\ln(1/8)/2 - (r' - r)/2, 0)$ is contained in $Q_{r'}$. Because $x < \ln(1/8)/2$, the Hessian matrix is not positive semidefinite at (x, y) . Therefore f is not convex on $Q_{r'}$. Thus r is maximally chosen.

Comment: Since the principal-minor criterion does not provide a statement for semidefinite matrices, it does not provide a statement about convexity of the function on the boundary of Q_r .

Exercise T11.3. The Hedge Algorithm and the Learning Rate

Problem

Consider an online prediction problem with $K = 2$ experts. Initialize the weights uniformly as $w_1^{(1)} = 1$ and $w_2^{(1)} = 1$. The observed loss vector in round 1 is $L^{(1)} = (0.2, 1.0)^T$.

- Let the learning rate be $\eta = 1$. State the multiplicative weight update rule and compute the updated probabilities $p_1^{(2)}$ and $p_2^{(2)}$ for the second round.
- The expected loss of the algorithm in any round t is defined as $\mathbb{E}[L^{(t)}] = \sum_{i=1}^K p_i^{(t)} L_i^{(t)}$. Compute the expected loss of the algorithm in round 1. How does it compare to the loss of the best expert in round 1?
- Consider the general update rule parameterized by $\eta > 0$. Analyze the mathematical limits of the updated probability distribution $p^{(2)}$ as $\eta \rightarrow 0$ and as $\eta \rightarrow \infty$. Discuss what these limits imply about the algorithm's behavior regarding exploration versus exploitation.

Solution

- The update rule for unnormalized weights is $w_i^{(t+1)} = w_i^{(t)} \exp(-\eta L_i^{(t)})$ with $p_i^{(t)} = \frac{w_i^{(t)}}{\sum_j w_j^{(t)}}$. With $\eta = 1$:

$$\begin{aligned} w_1^{(2)} &= e^{-0.2} \approx 0.818 \\ w_2^{(2)} &= e^{-1.0} \approx 0.368 \end{aligned}$$

Summing yields $W^{(2)} \approx 1.186$, resulting in:

$$p_1^{(2)} = \frac{e^{-0.2}}{e^{-0.2} + e^{-1.0}} \approx 0.690 \quad \text{and} \quad p_2^{(2)} = \frac{e^{-1.0}}{e^{-0.2} + e^{-1.0}} \approx 0.310$$

- In round 1, $p_1^{(1)} = p_2^{(1)} = 0.5$. The expected loss is:

$$\mathbb{E}[L^{(1)}] = 0.5(0.2) + 0.5(1.0) = 0.6$$

The best expert in hindsight has a loss of 0.2. The algorithm incurs a higher expected loss ($0.6 > 0.2$) because it initially lacks information and must distribute its trust equally to explore both options.

- The parameterized probabilities are $p_1^{(2)} = \frac{e^{-0.2\eta}}{e^{-0.2\eta} + e^{-1.0\eta}}$ and $p_2^{(2)} = \frac{e^{-1.0\eta}}{e^{-0.2\eta} + e^{-1.0\eta}}$.
 - Limit as $\eta \rightarrow 0$:** $\lim_{\eta \rightarrow 0} e^{-\eta L_i} = 1 \implies p^{(2)} \rightarrow (0.5, 0.5)^T$. *Implication:* The algorithm ignores performance differences completely, leading to **pure exploration** (random guessing).

- **Limit as $\eta \rightarrow \infty$:** Rewriting $p_1^{(2)} = \frac{1}{1+e^{-0.8\eta}}$. As $\eta \rightarrow \infty$, $e^{-0.8\eta} \rightarrow 0 \implies p_1^{(2)} \rightarrow 1$ and $p_2^{(2)} \rightarrow 0$. *Implication:* The algorithm shifts all mass to the historical top performer, matching the deterministic Follow-The-Leader strategy (**pure exploitation**).

Conclusion: A finite $\eta > 0$ balances this exploration-exploitation trade-off to minimize overall regret bounds.